

Tej Parekh

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EDUCATION

Brown University
Major: Computer Engineering, Degree: B.S.

- GPA: 4.0/4.0
- Coursework:** Computer Systems, Computer Networks, Circuits & Signals, E&M, Deep Learning, Linear Algebra, Probability & Statistics with Calculus, Data Structures & Algorithms, Differential Equations, Discrete Mathematics

May 2028

INTERNSHIP EXPERIENCE

MITRE (U.S. Secret Clearance Pending)
Signal Processing Intern

- Implementing Bayesian Kalman filter-like state estimation algorithms and simulations for a radar system (details restricted)
- Evaluating estimator robustness over noise models, initialization conditions, system assumptions, and simulation scenarios
- Received return offer for Summer 2027

May 2026 - August 2026
Bedford, MA

Portrait
Data Engineering Intern

- Built production AWS Glue ETL pipelines to ingest, transform, validate, and organize more than 100,000 electronic health record and order management records
- Wrote SQL queries in Athena supporting analytics dashboards and recurring business reporting in QuickSight

Jun. 2025 – Sep. 2025
San Diego, CA (Remote)

TECHNICAL LEADERSHIP, RESEARCH, AND PROJECTS

Brown Space Engineering
Control Systems Hardware Lead

- Leading hardware development for Attitude Determination and Control Systems of a CubeSat launching via SpaceX
- Implemented real-time ADCS control loops in MATLAB/Simulink, ported them to C for on-orbit autonomous operation
- Designed a custom photodiode sun sensor, characterized angular accuracy, and achieved ~60× commercial cost reduction

Aug. 2024 – Present
Providence, RI

RFC-Compliant TCP/IP Stack

- Implemented a userspace TCP/IP stack in Go, including connection establishment and teardown, sliding-window flow control, out-of-order buffering, retransmission queues, RTO-based timeouts, and zero-window probing

STM32 IMU Orientation and Sensor-Fusion Platform

- Built an embedded real-time STM32 six-axis IMU orientation estimation system in C with custom I²C drivers, 100 Hz timer-driven sampling, calibration/noise analysis, and adaptive Kalman and Mahony filtering

Brown Equitable Climate Futures Research Group
Undergraduate Researcher

- Developed a Python and GIS model integrating building geometry, satellite, demographic, and infrastructure data to estimate population across more than 400,000 Rhode Island buildings
- Distributed census population using multicriteria optimization, validation experiments, and probabilistic spatial allocation

May 2025 – Sep. 2025
Providence, RI

AWARDS

- Regeneron STS Scholar:** awarded to the top 300 student researchers nationwide
- National Merit Finalist:** awarded to the top 1% of PSAT scorers

Jan. 2024
Feb. 2024

SKILLS

- Languages:** C, C++, Python, Go, MATLAB, Java, SQL
- Technologies:** Git, AWS(Glue, Athena, QuickSight), Pandas, PyTorch, NumPy, GIS, STM32, Simulink, SolidWorks
- Areas:** State Estimation & Controls, Signal Processing, Simulation & Modeling, Embedded Systems, Data Engineering